

Project Charter
Course Team Project Charter
CE/CS/SE 3354 Software Engineering

1. Project Summary

Project Title	Campus Safety Incident & Emergency Accountability System
Project Sponsor	CE/CS/SE 3354 / Dr. Barbara Maweu
Project Manager	Kourosh Torkaman Sohrabi
Start Date	8/24/2026
Target End Date	12/09/2026
Charter Date / Version	9/13/2026 - v1

2. Purpose / Business Case

Explain why this project exists and what problem it solves. In a course context, this is your project motivation.

Problem Statement

Safety on campus is an ever-present issue that affects students' lives and their learning. UTD students may come across an array of hazards and not have an efficient way to inform staff about the issue. This can lead to these hazards being ignored and causing greater problems. Students may also not feel safe walking alone to the various places on campus, especially at night, as well as there being no quick way to confirm a student's safety during emergencies.

Business / Academic Justification

Solving these issues is vital to the safety of UTD's students. Students feeling unsafe on campus can lead to them not going to class or choosing to go to a different school entirely. Emergencies and hazards on campus can directly lead to injuries or deaths, which would negatively impact UTD in many ways. Implementing more safety features helps protect UTD's reputation and student body. Unlike tools that treat hazard reporting, escorted walks, and emergency headcounts as separate systems, this project links them through one shared incident state, so an expired safe walk or a severe hazard report automatically drives the emergency response.

3. Project Objectives

List specific, measurable objectives (align with your anticipated functional requirements).

- Objective 1 – Enable a student to submit a hazard report with a required category, location, and description, and receive a confirmation with a tracking status within the same session.

- Objective 2 – Enable authorized safety staff to advance a report through submitted → under review → resolved → closed, where a report cannot reach resolved without a recorded resolution note.
- Objective 3 – Enable students to check in during an active emergency event, with accounted-for and not-accounted-for counts updating within 5 seconds of a check-in.

4. Scope

In Scope

- User accounts with three distinct roles, student, safety admin, and floor warden, with role-based access controls limiting each role to its permitted actions and views.
- Hazard report submission requiring a category, location (building, then room/area), and description before submission, with students able to track the status of reports they've filed.
- A safety admin report queue enforcing the full status workflow (submitted → under review → resolved → closed), where a report cannot reach resolved without a recorded resolution note.
- A safe walk timer supporting one active session per student, a single time extension, and active/arrived/expired/cancelled states, where an expired session automatically generates a simulated notification and opens a new incident in the admin queue.
- An emergency check-in event feature allowing a floor warden to open an event for a building, prompting students to confirm their safety, and displaying a live accounted-for/not-accounted-for count that updates within 5 seconds of a check-in.

Out of Scope

- Integration with real 911, police, or emergency-service dispatch systems; the system only simulates these responses internally.
- Background or continuous GPS/location tracking of any user; only a stated destination or hazard location is recorded, never live position.
- Real-world notification delivery via SMS, push, or email; all alerts (e.g., a safe walk expiration) are simulated in-app rather than actually sent.

5. Stakeholders & Project Team

Identify everyone with a stake in the project and each team member's role.

Name	Role	Responsibility
Rajas Danekar	Emergency Muster Developer	Designs, implements, and tests the emergency check-in event feature, including event open/close logic and the live accounted-for headcount board.

Quan Do	Backend Developer	Managing reported databases from the user and implementing user's needed features. Responsible for maintaining each function and making sure the connection between three features works properly.
Quinn Merida Friman	Administration Manager	Designs, implements, and tests the account and access-control layer, including the login flow, the three user roles, and the permission rules that gate each role's actions and views, from requirements through test cases.
Neelesh Polireddy	Safe Walk Timer Developer	Designs, implements, and tests the safe walk timer feature, including session tracking, time extensions, autonomous timer expiration logic, and simulated alert triggering.
Kourosh Torkaman Sohrabi	Project Manager	Coordinates the project schedule, internal deadlines, and document assembly, owns the system architecture and the cross-component rules linking reporting, safe walk, and emergency events, and delivers the architecture section of each deliverable.
Nihal Khan	Student Side Developer	This role handles the student view by designing and implementing the student dashboard with report form, validation rules, and "my reports" view.
Nuzhat Tabassum	Admin Side Developer	This role handles the admin view by designing and implementing queue, status transitions, and resolution notes.

6. Deliverables & Milestones

High-level timeline of what will be produced and when.

Milestone / Deliverable	Owner	Target Date
Project Charter submitted	<i>Kourosh Torkaman Sohrabi</i>	<i>09/13/2026</i>
Deliverable #1: Requirements & diagrams finalized	<i>Nihal Khan / Nuzhat Tabassum</i>	<i>10/04/2026</i>
Deliverable #1: Architecture & design complete	<i>Quinn Merida Friman / Quan Do</i>	<i>10/25/2026</i>
Deliverable #2: Scheduling, Prototype implementation / Test Plan complete	<i>Rajas Danekar / Neelesh Polireddy</i>	<i>11/15/2026</i>
Deliverable #2: Final report & presentation complete	<i>All team members</i>	<i>12/06/2026</i>

7. Assumptions & Constraints

Assumptions

- There is a workforce (floor warden, safety admin, etc.) ready to be committed to monitoring safety alerts
- We assume no existing campus system already provides integrated hazard reporting and emergency accountability, so this project is not required to interoperate with one.
- Student can easily access reporting tool through a phone
- Drills practicing using the safety tool can be implemented

Constraints

- Limited scope to the UT Dallas Campus area
- No real emergency-service integration
- No medical records or HIPAA-applicable documents stored
- No continuous location tracing
- Notifications/Alerts delivery (SMS, push or email) will be simulated

8. Risks

Identify what could threaten the project's success and how you'll respond.

Risk	Likelihood	Impact	Mitigation
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Integration between three features	Medium	High	Define the share rules, especially on how safe walks create incident and emergency events affect active walks. Test the three connections before release
Exposure of sensitive user location data , a stated Safe Walk destination or a reported hazard location	Medium	High	Collect the minimum location data required, restrict read access by role so only authorized safety staff can view a destination or report location, and store no continuous position data.
False hazard report or location	Medium	Medium	Require category, location, and description before submission; tie every report to an authenticated account so submissions are attributable; and require staff severity review before a report can escalate to an emergency event, with the ability to close an invalid report with a recorded resolution note.

9. Budget & Resource Estimate

Hardware: The hardware cost will be \$0. The team will implement the bring your own device strategy.

Software: The software will cost \$0 because the team will utilize free tiers of cloud platforms. For example, we can use Firebase or Supabase for the database and core backend, WebSockets or Firestore listeners for the real time sync engine, or OAuth or Identity Provider for the user authentication module. The team needs to be mindful of the resource limits.

Personnel: This project will use 7 developers to create the product.

10. Success Criteria

Define what "done" and "successful" look like for this project.

- By the end of the project term, all three subsystems are fully specified and at least one is implemented as a working prototype, passing the test cases written for every rule in its state model, including the three cross-subsystem rules.
- By the end of the project term, users should rate the ease of navigating the tool during an emergency to be at least a 3/5.

11. Approval

All Team members sign-off indicates agreement on scope, roles, and objectives before work begins.

Each Team Member Name	Role	Signature / Date
Quan Do	Backend Developer	Quan Do/09-12-26
Quinn Merida Friman	Administration Manager	Quinn Merida Friman/09-12-26
Neelesh Polireddy	Safe Walk Timer Developer	Neelesh Polireddy / 9-13-26
Kourosh Torkaman Sohrabi	Project Manager	Kourosh Torkaman Sohrabi/09-12-26
Nihal Khan	Student Side Developer	Nihal Khan / 09-12-26
Nuzhat Tabassum	Admin Side Developer	Nuzhat Tabassum / 9-12-26
Rajas Danekar	Emergency Muster Developer	Rajas Danekar / 9-13-26

